

Presenter Checklist — Hands-On: Running QAOA on IBM Quantum Hardware

Part 01 · Knapsack QAOA · ~30 min · notebooks: [part01/knapsack_qaoa_guided.ipynb](#) → [ibm_hardware_qaoa_guided.ipynb](#)

① BEFORE YOU START (do this ~10 min early)

- **.env ready** — IBM_QUANTUM_TOKEN + IBM_QUANTUM_INSTANCE filled in; test load_dotenv prints True.

■ **Kernel warm** — run import qiskit and the imports cell once so the first live cell isn't slow.

■ **SUBMIT the hardware job EARLY** — open ibm_hardware_qaoa_guided.ipynb, set SUBMIT_HARDWARE_JOB=True, run it, and copy the printed job id. Queues are unpredictable — start it before the talk.
- **Save the job id** into JOB_ID_TO_MONITOR so you can poll it live. A fallback id is already in the cell.

■ **Pick a backend** — let Least_busy choose, or set IBM_BACKEND for a known-good device.

■ **Zoom / font** — bump Jupyter font size; pre-scroll to the first cell. Have slides on the second screen.

② RUN ORDER & TIMING (~30 min)

Min	Notebook / slide	Do / say	Watch for
0–3	Slides 1–2	Frame it: model → solve → access → run. Two ground rules (heuristic; opt-in).	everyone has the repo + .env open
3–9	knapsack_guided: instance → QUBO → Ising	Run cells; show the 6→10 variable jump and the exact optimum = 50.	variable count prints 10
9–14	knapsack_guided: QAOA + optimise + decode	Draw circuit; run COBYLA (maxiter 300 ≈ quick); sample; compare to 50.	energy falls then flattens
14–18	ibm_guided §1–4 + Slides 7–9	Load creds; connect; pre-flight checks. Reuse the job you submitted early.	token loaded = True; QPU reachable
18–24	ibm_guided §7–9 + Slides 10–11	Transpile (depth grows); per-qubit error map; layout matters.	depth jump; red = used qubits
24–28	ibm_guided §11–12 + Slides 12–13	Monitor your early job → DONE; compare sim vs hardware feasible rate.	QUEUED→RUNNING→DONE
28–30	Slide 14 + Q&A	Recap the recipe; point to VQE/TSP and Part 02. Take questions.	keep the job id

③ WHAT I CHANGED (safe to trust)④ IF SOMETHING GOES WRONG

- Guided copies made; originals untouched. Present from *_guided.ipynb.

Hardware is safe-by-default: SUBMIT_HARDWARE_JOB=False, MONITOR_JOB=False — a stray Run-All won't submit. Flip them yourself when ready.

Intro text fixed: the non-existent RUN_HARDWARE flag replaced with the real USE_IBM_BACKEND/SUBMIT_HARDWARE_JOB.

Knapsack COBYLA maxiter 1000→300 for a snappy live run (raise for tighter convergence).

Fixed “reps=1” wording to match the actual reps=2. Heavy outputs cleared (4.4 MB → 32 KB).
- If the queue is long: don't wait live — reconnect to the job you submitted early via service.job(job_id); the monitor cell has a fallback id so it never stalls.

If .env fails / no token: stay on the simulator — everything through §4 works; show transpilation/error-map read-only if a shared backend is reachable.

If a backend is retired/busy: set IBM_BACKEND to another, or drop least_busy.

Don't report the single most-frequent bitstring — keep the best feasible sample and state the feasible-shot rate.

Golden line: “On hardware, noise is part of the result — we measure it and report it.”